T2-25-26-AIT 846

Multi-Objective Machine Learning

Project report

Github

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Application: Fairness in Machine Learning

Dataset: Adult Census Income

Link

Date: 28th April, 2026

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Problem Setup

This project addresses fairness aware classification using the Adult Census Income dataset, where we formulate the task as a binary classification problem to predict whether an individual earns more than $50K annually.

A key concern in such predictive systems is bias with respect to protected attributes, particularly gender and race. Decisions influenced by biased models can lead to unfair outcomes across demographic groups. Therefore, this work formulates the problem as a multi-objective optimization (MOO) task, aiming to balance predictive performance with fairness.Bias is evaluated with respect to two protected attributes :

- Gender (binary: male/female)
- Race (multi-group, grouped to avoid sparsity issues)

The objective is to analyze trade-offs between accuracy and fairness and identify models that achieve an effective balance between these criteria.

Dataset & Preprocessing

The Adult Census Income dataset is used, containing demographic and employment-related attributes. The target variable indicates whether an individual earns more than $50K annually.

Data Cleaning and Preparation:

- Missing values were treated as NaN and removed.
- The target variable was cleaned and converted into a binary label as this (1: >50K, 0: $\leq$50K).
- Sensitive attributes considered:
  - Sex: encoded as a binary variable (Male = 1, Female = 0)
  - Race: low-frequency categories grouped into an "Other" class

Feature Engineering:

- Sensitive attributes were excluded from the feature set.
- Categorical features were encoded using one-hot encoding.
- Numerical features were standardized using z-score normalization.

Data Splitting:

- The dataset was split into training and testing sets (80/20) using stratified sampling to preserve class distribution.

Decision Variables

The optimization framework jointly tunes model parameters and fairness controls to explore trade-offs between predictive performance, fairness, and complexity.

Model-related variables:

- *Model type*: Logistic Regression or Random Forest
- *Regularization strength* (for Logistic Regression)
- *Number of trees* and *maximum depth* (for Random Forest)
- *Minimum samples per leaf*
- *Class weight scaling* to address class imbalance

Fairness-related variables:

- *Group-specific decision thresholds*: Separate thresholds are learned for each sensitive group (e.g., male/female), instead of a single global threshold. This enables post-processing control over prediction rates and improves fairness across groups.

Additional evaluation component:

- *Model complexity*: Incorporated as an objective to penalize overly complex models. For Logistic Regression, it is based on coefficient magnitude, while for Random Forest, it is a (log-scaled) function of total tree nodes.

All variables are evaluated using stratified cross-validation to ensure robust estimation, while avoiding degenerate solutions (e.g., extreme prediction rates). This setup allows the optimizer to effectively balance accuracy, fairness, and model simplicity.

Objective Selection

A key step in this work is selecting a compact and non-redundant set of objectives for multi-objective optimization. The following candidate objectives were initially considered:

- Predictive performance (F1-score)
- Demographic Parity (DP)
- Equalized Odds (EO)
- Individual Fairness (IF)
- Model Complexity

Initial Analysis:

Preliminary experiments indicate strong dependencies among the fairness metrics when *sex* is used as the protected attribute. In particular:

- Demographic Parity (DP) and Equalized Odds (EO) exhibit a strong positive correlation.
- Individual Fairness (IF), computed using feature-space similarity, also shows a close alignment with group fairness metrics.

This behavior is confirmed both numerically and visually through the threshold-based trade-off analysis plots (F1 vs DP, F1 vs EO, and F1 vs IF). The plots show highly similar trends, indicating that these fairness notions do not provide independent optimization directions in this dataset.

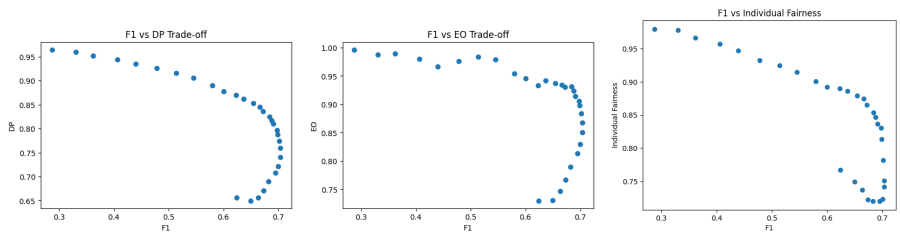

Figure 1: Trade-off analysis: F1 vs DP, EO, and Individual Fairness

This is further supported by the **objective correlation matrix**:

	F1	**DP**	**EO**	**IF**
F1	1.00	−0.75	−0.59	−0.80
DP	−0.75	1.00	0.97	0.97
EO	−0.59	0.97	1.00	0.91
IF	−0.80	0.97	0.91	1.00

The strong correlations, along with similar patterns observed in the trade-off plots, lead to redundancy in the objective space and reduce the diversity of Pareto-optimal solutions.

Final Objective Set:
Based on this analysis, the final selected objectives are:

- Maximize F1-score (predictive performance)
- Maximize Demographic Parity (fairness)
- Minimize Model Complexity

Justification:

- **F1-score**: captures precision-recall balance and is suitable for imbalanced classification tasks.
- **Demographic Parity**: chosen as the primary fairness metric to enforce equal positive prediction rates across groups.
- **Model Complexity**: included to encourage simpler and more interpretable models.

Metrics

The optimization process is guided by three objectives: predictive performance, fairness, and model complexity. Each is quantified using the following metrics.

1. F1-score (Predictive Performance): The F1-score is used to measure the balance between precision and recall in binary classification.

```
f1 = f1_score(y_true, y_pred)
```

2. Demographic Parity (Fairness): Fairness is quantified using Demographic Parity difference, converted into a maximization objective.

```
dp = 1 - abs(
    demographic_parity_difference(
        y_true, y_pred,
        sensitive_features=sensitive
    )
)
```

3. Model Complexity: Model complexity is included as a regularization objective to encourage simpler and more interpretable models. It is defined differently depending on the model type and is subsequently log-scaled to ensure comparability across model families.

For Logistic Regression, complexity is measured as the sum of absolute coefficient values, while for Random Forests it is based on the total number of tree nodes. A logarithmic transformation is applied to prevent large-scale dominance and normalize the measure:

```
if model_type == 0:
    complexity = np.sum(np.abs(model.coef_))
else:
    complexity = np.log(sum(tree.tree_.node_count
                            for tree in model.estimators_) + 1)
```

The analysis highlights that fairness metrics are often not independent in practice. Careful selection of objectives is necessary to ensure meaningful Pareto fronts in multi-objective optimization.

Optimization Framework

The multi-objective optimization problem is solved using NSGA-II (Non-dominated Sorting Genetic Algorithm II), an evolutionary algorithm designed to efficiently handle multiple conflicting objectives.

Why NSGA-II: NSGA-II is chosen due to its ability to generate a well-distributed set of Pareto-optimal solutions while maintaining diversity using non-dominated sorting and crowding distance. It is particularly effective for problems involving trade-offs between predictive performance, fairness, and model complexity.

Optimization Pipeline: The optimization follows an iterative evolutionary process in which a population of candidate solutions is evolved over successive generations. Each solution encodes both model hyperparameters and fairness-related decision thresholds.

The process can be summarized as:

$$Initialization \rightarrow Evaluation \rightarrow Non-dominatedSorting \rightarrow Selection$$

$$\rightarrow Crossover\&Mutation \rightarrow Iteration$$

Evaluation Strategy: Each candidate solution is evaluated using a robust cross-validated procedure with additional fairness-aware constraints:

- **Cross-validation:** A 3-fold stratified split is used to ensure stable estimates of performance and fairness.
- **Group-specific thresholds:** Separate decision thresholds are learned for each sensitive group, enabling finer fairness control.
- **Degeneracy filtering:** Solutions producing extreme prediction rates (very low or very high positive rates) are discarded to avoid trivial or unstable models.

Summary: NSGA-II enables systematic exploration of the trade-off space between F1-score, fairness, and model complexity, producing a diverse Pareto front of candidate solutions.

To operationalize this framework, the optimization problem is encoded in a structured form where each individual in the population represents a combination of model hyperparameters and fairness-related threshold parameters. This formulation allows the evolutionary process to directly search over both predictive and fairness-related decision spaces in a unified manner.

Objective Formulation in PyMOO: The optimization problem is implemented using the PyMOO framework as follows:

```
class FairnessMOO(Problem):

    def __init__(self):
        super().__init__(
            n_var=8,
            n_obj=3,
            xl=np.array([0, 1e-4, 100, 5, 1, 0.5, 0.3, 0.3]),
            xu=np.array([1, 10, 500, 25, 10, 3.0, 0.7, 0.7])
        )

    def _evaluate(self, X, out, *args, **kwargs):

        results = []

        for x in X:
            f1, dp, comp = evaluate_solution(x)

            results.append([
                -f1,   # maximize F1
                -dp,   # maximize fairness
                comp   # minimize complexity
            ])

        out["F"] = np.array(results)
```

Results : Quantitative and Qualitative Analysis

This section presents a detailed quantitative evaluation of the obtained optimization results. We first analyze the full solution space and the resulting Pareto front with *gender* as the protected attribute, followed by a comparative analysis with *race* based fairness to understand how the choice of sensitive attribute affects the observed trade offs.

Global vs Pareto Solution Space (Gender)

To evaluate the effectiveness of the optimization process, we compare the full set of evaluated solutions with the extracted Pareto-optimal front.

At each generation, NSGA-II maintains a population of **100 candidate solutions** (pop_size = 100). After the final generation, the algorithm returns the last population, which represents the best evolved set of solutions. These 100 points correspond to the **final generation's population**, not all historical evaluations.

From this population, non-dominated sorting is applied to identify Pareto-optimal solutions. In our case, **19 out of 100 solutions** are found to be non-dominated and form the **Pareto front**.

Figure 2 visualizes this process.

- The **left plot** shows all 100 solutions in the final population (global view).
- The **middle plot** highlights only the Pareto-optimal solutions (local optimal boundary).
- The **right plot** overlays both, clearly showing how dominated solutions are filtered out.

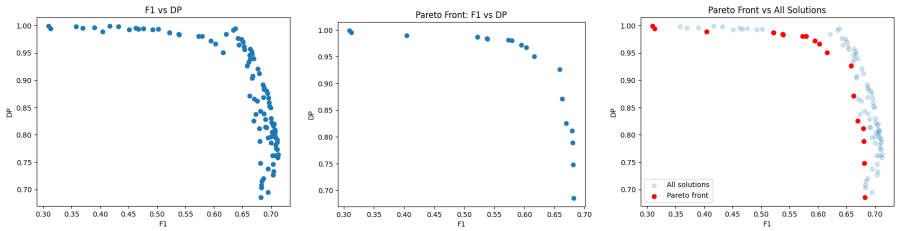

Figure 2: Global vs Pareto solution space for gender: (left) all solutions in the final population (100 points), (middle) extracted Pareto front (19 non-dominated solutions), (right) comparison showing dominated vs non-dominated solutions in the F1–DP objective space.

Key observations:

- The full population forms a dense cloud, indicating that many solutions are sub-optimal (dominated).
- The Pareto front lies along the upper boundary, where no solution can improve one objective without worsening another.
- Only **19% of solutions (19/100)** are Pareto-optimal, showing that the majority of the search space consists of inferior trade-offs.

This distinction between the global solution space (all evaluated candidates in the final generation) and the Pareto front (optimal trade-offs) demonstrates that NSGA-II effectively filters and preserves only the most competitive solutions.

Fairness trade-off (Gender)

The Pareto front for gender-based fairness is shown in Figure 3, illustrating the trade-off between predictive performance (F1-score) and fairness (Demographic Parity).

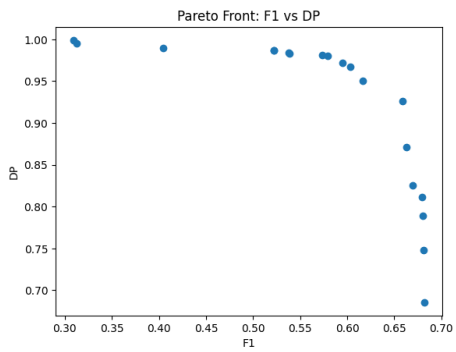

Figure 3: Pareto front for gender showing the trade-off between F1-score and Demographic Parity (DP).

A clear inverse relationship is observed between the two objectives: as F1-score increases, demographic parity decreases. The observed ranges are:

- F1-score: approximately 0.30 to 0.71
- Demographic Parity: approximately 0.68 to 1.00

Key observations:

- High-performance models (F1 $\approx$ 0.68–0.71) achieve only moderate fairness (DP $\approx$ 0.75–0.85).
- Near-perfect fairness (DP $\rightarrow$ 1.0) is achievable only at significantly lower performance levels (F1 $\approx$ 0.30–0.50).
- The transition between these regions is gradual, indicating a continuous trade-off rather than abrupt changes.

The Pareto front exhibits a smooth, convex shape, which suggests that:

- The objectives are strongly conflicting but well-structured
- NSGA-II has effectively converged to a stable set of non-dominated solutions

This behavior indicates that improving fairness consistently requires sacrificing predictive performance and vice versa.

Complexity Trade-off (Gender)

To analyze the relationship between predictive performance and model simplicity, Figure 4 presents the Pareto front in the F1-score vs model complexity space.

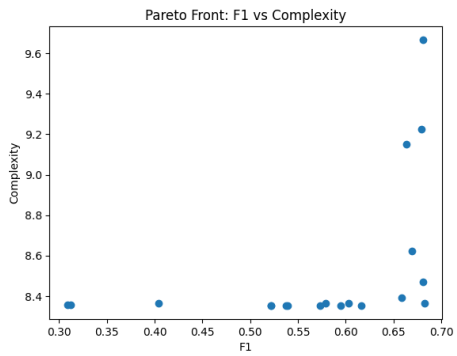

Figure 4: Pareto front for gender showing the trade-off between F1-score and model complexity (log-scaled).

The observed complexity values (log-scaled) range from approximately 8.3 to 14.5, capturing both simple and highly complex models.

Key observations:

- High-performance solutions (F1 $\approx$ 0.65–0.70) are generally associated with higher complexity, indicating the use of more expressive models.
- Low-complexity models (complexity $\approx$ 8.3–9.0) tend to achieve either lower predictive performance or moderate performance with improved fairness balance.
- A cluster of solutions exists in the mid-range (F1 $\approx$ 0.55–0.65) with relatively low complexity, representing efficient trade-offs.

Overall, this reveals a secondary trade-off in the optimization process:

> Improving predictive performance often requires increased model complexity.

This highlights the importance of including model complexity as an explicit objective, as it enables the identification of solutions that balance accuracy, fairness, and efficiency.

3D Pareto Structure (Gender)

To better understand the joint interaction between all three objectives, Figure 5 presents the Pareto front in a three-dimensional space defined by F1-score, Demographic Parity (DP) and model complexity.

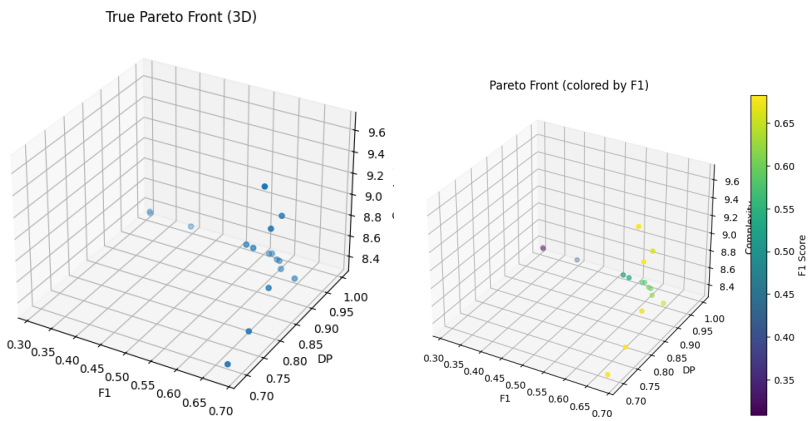

Figure 5: 3D Pareto front for gender: (left) Pareto solutions in F1–DP–Complexity space, (right) the same front colored by F1-score to highlight performance variation across the surface.

The Pareto solutions form a curved surface rather than a linear structure, indicating that all three objectives are simultaneously conflicting. This confirms that the problem cannot be decomposed into simpler pairwise trade-offs without loss of information.

Key observations:

- Solutions with high fairness (DP $\approx$ 0.95–1.0) and low complexity are associated with lower predictive performance (F1 $\approx$ 0.30–0.50).
- High-performance solutions (F1 $\approx$ 0.65–0.70) generally require higher complexity and exhibit moderate fairness levels.
- The colored visualization shows a smooth gradient, indicating a continuous transition of performance across the Pareto surface rather than isolated clusters.

Insight:

> The presence of a structured Pareto surface confirms that optimizing any single objective (performance, fairness, or complexity) inevitably impacts the others, validating the need for a full multi-objective optimization framework.

Overall, this visualization highlights the multi-dimensional nature of the trade-offs, where different regions of the Pareto surface correspond to distinct modeling strategies ranging from fairness oriented to performance driven solutions.

Knee Point and Candidate Selection (Gender)

To identify practically useful solutions from the Pareto front, we perform a *knee point analysis*, which selects solutions closest to an ideal trade-off across all objectives. Top 3 candidate solutions are shown in Table 1

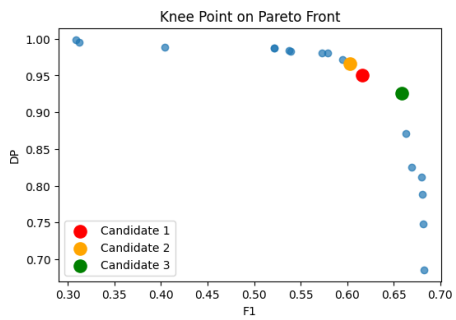

Figure 6: Knee point visualization on the Pareto front (F1 vs DP) for gender. The top 3 candidate solutions are highlighted based on proximity to the ideal point.

Methodology:

- All Pareto-optimal solutions are first normalized to the range $[0, 1]$ to ensure comparability across objectives.

- An **ideal reference point is then defined as (1,1,0)** representing maximum F1-score, maximum fairness and minimum model complexity.

- For each solution on the Pareto front, the Euclidean distance to this ideal point is computed, and the solutions are ranked accordingly.

- A smaller distance indicates a better overall trade-off between performance, fairness, and complexity.

Top 3 Candidate Solutions:

Observations:

- **Candidate 1 (knee point)** provides the best overall balance between performance, fairness, and complexity.

Table 1: Top representative candidate solutions from the Pareto front showing trade-offs between F1-score, Demographic Parity, and model complexity.

Candidate	F1-score	DP	Complexity	Interpretation
C1	0.616	0.950	8.35	Balanced (knee point)
C2	0.603	0.967	8.37	Fairness-oriented
C3	0.659	0.926	8.39	Performance-oriented

- **Candidate 2** achieves higher fairness with only a small drop in F1, making it suitable for fairness-critical applications.
- **Candidate 3** improves performance but sacrifices fairness, indicating diminishing returns beyond the knee region.

Beyond the knee point, small gains in F1-score result in disproportionately large reductions in fairness, highlighting the importance of selecting solutions near the knee for practical deployment.

This analysis provides a principled way to select a single solution from the Pareto front based on application-specific priorities.

Pareto Front Analysis (Race)

To maintain consistency, we perform the same analysis using *race* as the sensitive attribute. Figure 7 presents the global solution space and the extracted Pareto front.

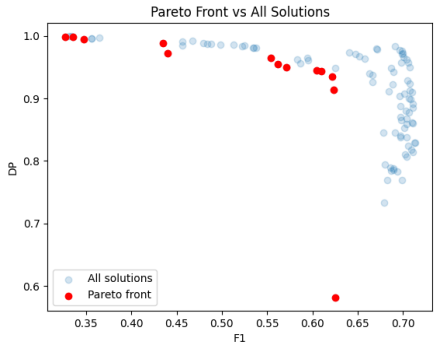

Figure 7: Global vs Pareto solution space for race: dominated solutions (blue) vs non-dominated Pareto front (red). A total of 13 Pareto-optimal solutions are identified from the final population.

Compared to gender, the Pareto front for race is more compact, with only 13 non-dominated solutions, indicating a more constrained solution space.

For completeness, we also visualize the Pareto trade-offs across all objectives.

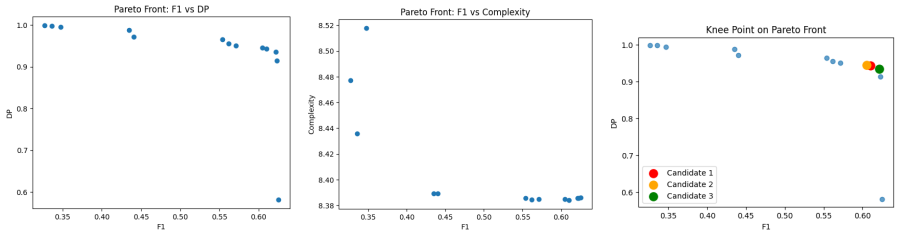

Figure 8: Pareto analysis for race: (left) F1 vs DP trade-off, (middle) F1 vs complexity, (right) knee point and top candidate solutions.

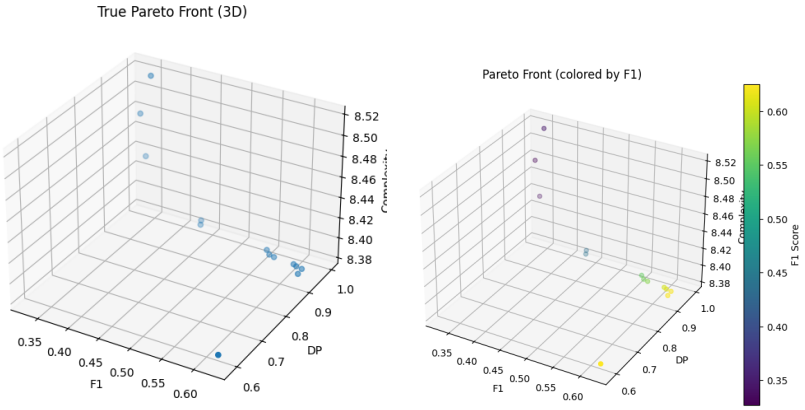

Figure 9: 3D Pareto front for race: (left) F1–DP–Complexity space, (right) colored by F1-score to show performance distribution.

The top candidate solutions identified via knee-point analysis:

Top 3 candidate solutions:

Candidate	F1	DP	Complexity	Interpretation
C1	0.610	0.943	8.38	Balanced (knee point)
C2	0.605	0.945	8.38	Fairness-oriented
C3	0.622	0.935	8.39	Performance-oriented

A similar analysis as performed for gender can be applied here; however, rather than repeating detailed interpretations, we now focus on comparing how the optimization behavior changes when the sensitive attribute shifts from gender to race.

Comparison of Gender vs Race Fairness

To compare the optimization behavior under different sensitive attributes, we analyze the resulting Pareto fronts using two standard multi-objective indicators: *hypervolume* and *spacing*.

Hypervolume (HV) measures the volume of the objective space dominated by the Pareto front with respect to a reference point. In this work, all objectives are first normalized to $[0, 1]$, and a reference point $(1.1, 1.1, 1.1)$ is used, slightly worse than the normalized objective space. A higher hypervolume indicates better convergence towards the ideal region and a stronger overall Pareto front.

Spacing measures the uniformity of distribution of solutions along the Pareto front. It is computed as the standard deviation of the minimum pairwise distances between solutions; lower values indicate a more evenly distributed front.

The computed results are as follows:

Sensitive Attribute	**Hypervolume**	**Spacing**
Gender	0.392	0.102
Race	0.531	0.084

- **Hypervolume:** The race-based setup achieves a higher hypervolume, indicating that its Pareto solutions are closer to the ideal region in the normalized objective space. This suggests stronger overall performance in terms of combined fairness, accuracy, and complexity.

- **Spacing:** Race also exhibits slightly lower spacing, indicating a more uniformly distributed Pareto front compared to gender.

It can be observed that:

- **Gender:** Produces a wider spread of solutions, including both high-performance and low-fairness regions, resulting in greater variability along the Pareto front.

- **Race:** Produces a more concentrated Pareto front, with solutions clustered in a narrower region dominated by fairness constraints.

	Gender	
Solution	**F1-score**	**DP**
Best F1	0.6823	0.6856
Best DP	0.3088	0.9990

	Race	
Solution	**F1-score**	**DP**
Best F1	0.6252	0.5816
Best DP	0.3268	0.9989

Analysis:

The comparison between gender and race as sensitive attributes reveals consistent but structurally different trade-offs between predictive performance and fairness.

Across both settings, the *Best F1* solutions prioritize predictive accuracy at the cost of reduced fairness, while the *Best DP* solutions achieve near-perfect demographic parity but significantly degrade predictive performance. This confirms a strong inherent conflict between accuracy and fairness objectives in the dataset.

A key observation is that the magnitude of this trade-off differs across sensitive attributes. For gender, the Best F1 solution achieves higher predictive performance (F1 ≈ 0.68) while still maintaining moderate fairness (DP ≈ 0.69), whereas for race, the Best F1 solution exhibits noticeably lower fairness (DP ≈ 0.58), indicating a sharper degradation in fairness for comparable accuracy gains.

Conversely, both gender and race achieve near-perfect demographic parity in the Best DP solutions (DP ≈ 1.0), but at the cost of a substantial drop in F1-score (around 0.30–0.33). This indicates that enforcing strict fairness constraints leads to highly conservative classifiers with limited predictive utility.

Non-obvious trade-off: A key observation is that achieving similar levels of fairness does not guarantee similar predictive performance across different sensitive attributes. In particular, race-based fairness constraints appear to impose a stronger restriction on the hypothesis space, resulting in lower achievable F1-scores even at comparable fairness levels.

Quantitative trade-off (Gender):

1. Accuracy gain vs fairness loss: An increase in F1-score from approximately 0.60 to 0.68 corresponds to a decrease in demographic parity from about 0.95 to 0.75, indicating a fairness degradation of roughly 20%.

2. Fairness gain vs accuracy loss: Improving fairness from DP ≈ 0.75 to DP ≈ 0.98 leads to a reduction in F1-score from approximately 0.68 to 0.45, indicating a performance loss of over 30%.

3. Overall trade-off magnitude: This corresponds to a performance loss of over 30% when moving towards highly fair solutions, highlighting the steep nature of the fairness–accuracy trade-off.

Conclusion: NSGA-II effectively captures these trade-offs by generating a diverse Pareto front spanning both high-performance and high-fairness regimes. However, the results also highlight that the fairness–performance relationship is sensitive to the choice of protected attribute, which significantly influences the structure and spread of the solution space.

Tables 2 and 3 clearly demonstrate the Pareto trade-off structure learned by NSGA-II. The gender setting exhibits a smoother transition between performance and fairness, while the race setting shows stronger fairness constraints that more aggressively reduce predictive performance. This highlights how the choice of sensitive attribute significantly affects the shape and distribution of Pareto-optimal solutions.

Pareto Points Tabulation:

A tabulation of representative Pareto-optimal solutions is provided for both gender and race settings. These solutions are selected from the final non-dominated set and illustrate the trade-off structure between the chosen objectives.

- For the *gender*-based setting, the Pareto points show a gradual increase in fairness (DP) with a corresponding slight decrease in predictive performance (F1-score), while model complexity varies moderately.

- The *race*-based setting, in contrast exhibits stronger fairness constraints, where DP increases significantly but with a more noticeable reduction in F1-score, while complexity remains relatively stable.

Pareto Solutions (Gender):

Table 2: Pareto-optimal solutions for Gender showing F1-score, Demographic Parity, and Model Complexity.

F1-score	**DP**	**Complexity**
0.6823	0.6856	8.3646
0.6812	0.7484	8.4692
0.6805	0.7886	9.6677
0.6795	0.8117	9.2252
0.6694	0.8253	8.6223
0.6630	0.8714	9.1522

Pareto Solutions (Race):

Table 3: Pareto-optimal solutions for Race showing F1-score, Demographic Parity, and Model Complexity.

F1-score	**DP**	**Complexity**
0.6252	0.5816	8.3861
0.6231	0.9143	8.3855
0.6218	0.9349	8.3855
0.6101	0.9431	8.3841
0.6050	0.9450	8.3849
0.5713	0.9501	8.3847

Optimization Behavior and Convergence Analysis

The NSGA-II algorithm was executed with a **population size of 100 for 80 generations**. Image in Figure 10 shows progress for some initial generations. At each generation, the algorithm evaluates a new population and applies non-dominated sorting and diversity preservation to evolve the Pareto front.

n_gen	n_eval	n_nds	eps	indicator
1	100	32	-	-
2	200	37	0.0226061479	ideal
3	300	55	0.0237169803	ideal
4	400	72	0.6657448268	nadir
5	500	77	0.0044795269	ideal
6	600	80	0.0137304954	ideal
7	700	100	0.0093270110	ideal
8	800	100	0.0027458282	ideal
9	900	100	0.0039366119	ideal
10	1000	100	0.0047489210	ideal
11	1100	100	0.0090944216	f
12	1200	100	0.0105380266	f
13	1300	100	0.0072427826	ideal
14	1400	100	0.0449575033	nadir
15	1500	100	0.0086903149	f
16	1600	100	0.0163285553	f
17	1700	100	0.0088671260	f
18	1800	100	0.0666019156	nadir
19	1900	100	0.0098270386	f
20	2000	100	0.0041027272	nadir
21	2100	100	0.0025673461	nadir
22	2200	100	0.0076847115	f
23	2300	100	0.0956595912	nadir
24	2400	100	0.0202838264	nadir
25	2500	100	0.0732638680	nadir
26	2600	100	0.1075515074	nadir
27	2700	100	0.0049178862	f

Figure 10: Your caption describing the figure.

The convergence behavior can be analyzed using three key indicators:

1. Number of Non-Dominated Solutions (n_nds): The number of Pareto-optimal solutions increases rapidly in the initial generations (from 32 at generation 1 to 100 by generation 7), indicating fast exploration of the solution space. After this point, the population becomes fully non-dominated, suggesting that the algorithm has identified a stable Pareto front.

2. Epsilon Indicator (ϵ): The epsilon values measure improvement between successive generations. Early generations show relatively large improvements (e.g., $\epsilon \approx 0.66$ at generation 4), while later generations exhibit very small changes ($\epsilon < 0.02$), indicating convergence towards a stable front.

3. Ideal and Nadir Indicators: The optimization alternates between *ideal* and *nadir* improvements. - *Ideal* indicates improvement in best objective values (moving toward optimal solutions). - *Nadir* reflects refinement of worst-

case solutions on the Pareto front.

This alternating pattern suggests that NSGA-II is not only improving the best solutions but also refining the diversity and spread of the Pareto front.

Convergence Insight: After approximately 20–30 generations, improvements become marginal, and the Pareto front stabilizes. Subsequent generations primarily refine solution distribution rather than discover fundamentally new trade-offs.

Interpretation: The rapid increase in n_nds followed by stabilization, along with decreasing epsilon values, confirms that NSGA-II efficiently converges to a well-distributed Pareto front. However, the oscillations in the epsilon and indicator values highlight minor instability in later generations, reflecting sensitivity to stochastic operations such as mutation and crossover.

Overall, the algorithm demonstrates strong convergence properties while maintaining solution diversity, making it suitable for fairness-aware multi-objective optimization.

Notably, the early saturation of non-dominated solutions suggests that the problem admits a relatively dense Pareto front, making diversity preservation a critical component of the optimization process.